VID

Virtual Identification System

ERP-as-a-Service for Educational Institutions

Multi-Tenant · Modular · Scalable | 2026

Multi-Tenant

Architecture

10+ Core

Workspaces

AI-Powered

YANTRA Engine

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YANTRA Engine & Future Roadmap

Fragmented Systems Create Critical Operational Gaps

THE EDUCATION ERP GAP

Fragmented School Systems

Institutions run disconnected tools for admissions, academics, fees, and exams — no single source of truth.

Lack of Scalability

Legacy software cannot support multiple institutions or growing student bodies without performance collapse.

No Centralized Management

Each school operates in isolation with no unified oversight, reporting, or cross-campus analytics.

Manual Processes

Paper-based attendance, fee collection, and reporting drain resources and introduce human errors.

The Impact: Administrative overhead increases **40%+** while data visibility drops to near zero across campuses. Schools struggle with compliance, parents lack transparency, and management cannot make data-driven decisions.

Without a unified platform, educational institutions face rising costs, declining efficiency, and frustrated stakeholders at every level.

VID Unifies Education Management on a Single Scalable Platform

THE VID SOLUTION OVERVIEW

Unified Modular ERP

One platform integrating admissions, academics, fees, exams, and HR — all connected, all visible.

Multi-Tenant Architecture

One platform serving CBSE, ICSE, IGCSE institutions with complete data isolation between tenants.

Role-Based Access Control

Every user sees only what they need — from Principal to Student — with granular workspace permissions.

Workspace-Based System

Functionality organized into plug-and-play modules that institutions enable as needed — no bloat.

The Result: Centralized governance with full institution autonomy. Super admins maintain global control while each school self-manages day-to-day operations independently.

VID transforms scattered tools into a coherent ecosystem — reducing overhead, improving visibility, and empowering every stakeholder.

Centralized Governance Meets Institution Autonomy

PRODUCT OVERVIEW

CENTRALIZED GOVERNANCE

- Policy enforcement across all institutions
- Global reporting and cross-institution analytics
- Unified user management and security policies
- Central billing and subscription control

INSTITUTION AUTONOMY

- Independent admissions and enrollment workflows
- Custom fee structures and academic calendars
- Self-managed faculty and student records
- Workspace selection based on institutional needs

Key Insight: VID gives each institution the freedom to operate independently while providing management with the visibility and control they need — the best of both worlds.

Four Strategic Objectives Drive Platform Design

CORE OBJECTIVES

01

Multi-Institution

Support unlimited schools under one tenant-aware infrastructure with complete data separation.

02

Secure & Scalable

PostgreSQL with Row-Level Security, API-first backend, and horizontal scaling readiness.

03

Modular Extensibility

Add workspaces without disrupting existing operations — true plug-and-play architecture.

04

Role-Based Control

Granular permissions from Super Admin down to individual students and parents.

Design Philosophy: VID is architected around four non-negotiable principles that every feature, workspace, and API endpoint must satisfy.

Multi-Institution means one platform installation serves a network of schools — each with isolated data, custom branding, and independent configuration.

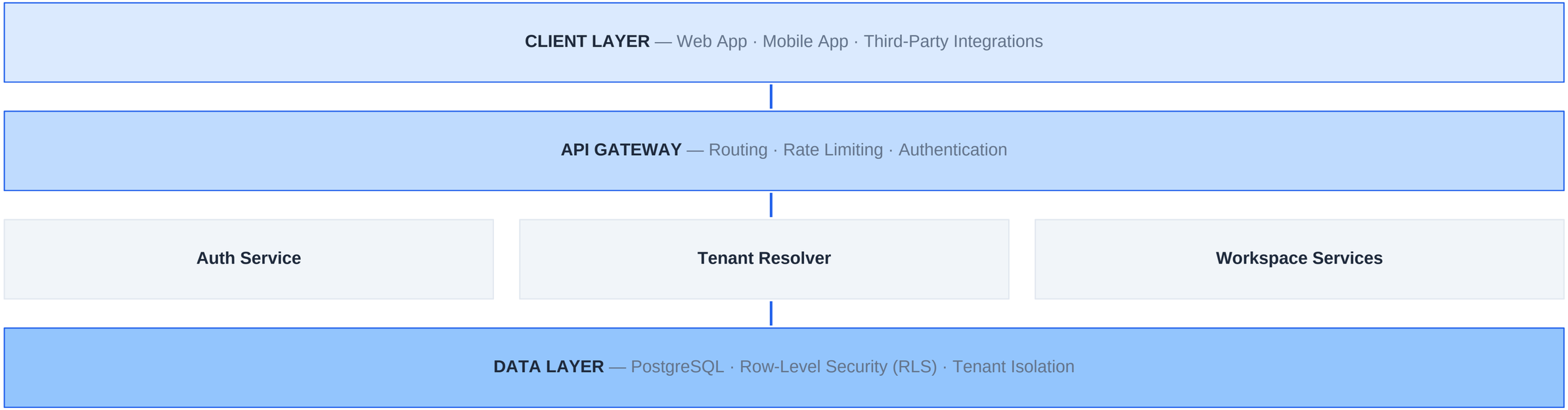
Security & Scalability are built in from the ground up — not bolted on. Row-Level Security, encrypted APIs, and containerized deployment ensure enterprise-grade protection.

Modularity guarantees that adding a new workspace (e.g., Transport) never requires downtime or affects existing modules.

RBAC ensures that a student cannot see faculty salaries, and an accountant cannot modify exam results — every action is authorized.

Multi-Tenant Architecture with Complete Data Isolation

SYSTEM ARCHITECTURE



Multi-Tenant Model: A single PostgreSQL database serves all institutions. Row-Level Security policies enforce tenant isolation at the database layer — ensuring School A can never access School B's data, even by accident.

API-Based Backend: Every workspace communicates through RESTful APIs. This enables third-party integrations, mobile applications, and future marketplace extensions without core platform changes.

Workspace-Based Design: Each module (Academic, Payments, HRMS) operates as an independent service. They share the API bus but maintain separate schemas — enabling true plug-and-play functionality.

Five-Level Hierarchy Ensures Controlled Access

HIERARCHICAL STRUCTURE

Super Admin — Platform control, tenant creation, global policies

Institutions — School config, branding, curriculum selection

Admin — User management, workspace assignment, oversight

Workspace Users — Faculty, Accountant, Role-specific access

End Users — Students and Parents with limited permissions

HIERARCHY PRINCIPLES

- Top-down policy enforcement
- Bottom-up data aggregation
- Role inheritance at each level
- Cross-level audit trails
- Permission scope narrowing

How it works: A Super Admin creates a tenant for "Greenwood Academy." The school's Admin assigns workspaces and invites Faculty. Faculty access only their assigned modules (e.g., Academic, Examination). Students see only their personal data — timetable, results, and fee status. Every level inherits permissions from above but cannot escalate beyond their scope.

Granular RBAC Defines Who Can Access What

ROLE-BASED ACCESS CONTROL

Principal

Full institution oversight, cross-workspace reporting, policy approval, and strategic decision-making access.

Faculty

Teaching workspace access, grade entry, attendance marking, timetable view, and student communication.

Accountant

Payments, fee collection, payroll processing, receipt generation, and financial reporting access.

Student

Personal records, timetable, exam results, fee status, and limited communication channels.

WORKSPACE-LEVEL PERMISSIONS

Each role maps to specific **CRUD** (Create, Read, Update, Delete) operations within each workspace module. Permissions are evaluated at three levels:

Platform Level: Super Admin controls tenant creation, billing, and global security policies.

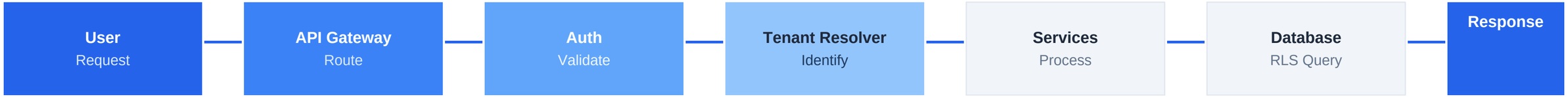
Institution Level: Principal and Admin manage users, workspaces, and institutional settings.

Workspace Level: Faculty can create grades in Examination but only view (not edit) Payments. Accountants can process fees but cannot modify academic records.

This granular control ensures **principle of least privilege** — every user has exactly the access they need, nothing more.

Every Request Flows Through a Secure, Resolved Pipeline

REQUEST FLOW ARCHITECTURE



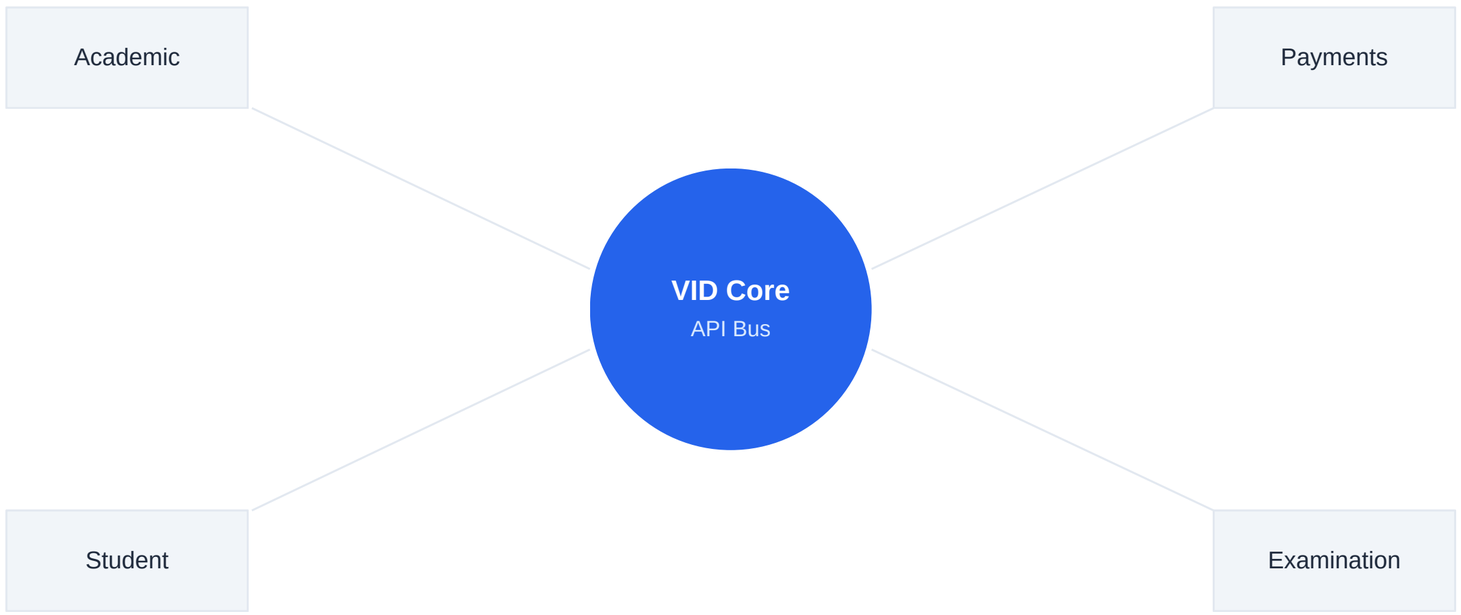
REQUEST LIFECYCLE

1. **User Request:** A teacher opens the Examination workspace and clicks "View Results." The frontend sends an authenticated API request.
2. **API Gateway:** The gateway receives the request, applies rate limiting, and forwards it to the appropriate service router.
3. **Auth Validation:** The JWT token is validated. The system confirms the user is active and has not exceeded their session timeout.
4. **Tenant Resolution:** The tenant ID is extracted from the token. All subsequent queries are scoped to this institution only.
5. **Service Processing:** The Examination service processes the request, applying workspace-specific business logic.
6. **Database Query with RLS:** PostgreSQL executes the query with Row-Level Security policies automatically appending WHERE tenant_id = ? — impossible to bypass.
7. **Filtered Response:** Only authorized, tenant-scoped data returns to the user. The entire pipeline completes in milliseconds.

> **Security Guarantee:** Even if a malicious actor bypasses the application layer and connects directly to the database, RLS policies at the PostgreSQL level prevent cross-tenant data access. Tenant isolation is **hardware-enforced**, not just application-logic enforced.

Workspaces — The Modular Building Blocks of VID

WORKSPACE ARCHITECTURE



WORKSPACE CONCEPT

- Self-contained functional modules
- Independent database schemas
- Dedicated API endpoints
- Isolated UI components
- No direct coupling between modules
- Subscribe only to needed workspaces

Workspace Philosophy: VID is an operating system; workspaces are applications. Institutions install only what they need — no bloat.

A primary school might use **Academic, Student, Faculty, Payments, and Timetable**. A multi-campus group adds **Transport, Library, and Parent Portal** — scaling without complexity.

Workspaces communicate via the central API bus with shared auth but independent schemas. **Updating Library never breaks Examination.**

10 Core Workspaces Power Daily School Operations

CORE WORKSPACES

Academic

Curriculum, subjects, syllabi, lesson planning

Student

Enrollment, profiles, documents, parent linkage

Faculty

Recruitment, assignments, performance, leave

Examination

Schedule, halls, seating, results, report cards

Admission

Applications, interviews, merit lists, enrollment

Payments

Fee structures, online collection, receipts, dues

HRMS

Payroll, attendance, claims, compliance

Timetable

Auto-generation, conflict detection, substitution

Common

Resources, notices, calendar, announcements

Principal

Cross-workspace analytics, KPIs, alerts

Core Workspace Design: These 10 workspaces form the foundation of every VID deployment. They cover the operational spectrum from student enrollment to exam results, from fee collection to staff payroll.

Each workspace is independently deployable and upgradable. A bug fix in **Timetable** never affects **Payments**. A new feature in **Academic** rolls out without requiring downtime in **Examination**.

The **Principal Dashboard** is unique — it aggregates data from all other workspaces, giving leadership a single-pane view of institutional health.

Recommended Workspaces Extend Operational Coverage

RECOMMENDED WORKSPACES

Transport

Route planning, vehicle tracking, fee integration, safety alerts

Library

Catalog, issue/return, fines, digital resources

Parent Portal

Real-time updates, fee view, communication, calendar

Events

Planning, budgeting, registration, feedback, gallery

Discipline

Incident logging, counseling, behavioral analytics

Sports

Team management, inventory, schedules, achievements

Health

Medical records, vaccination tracking, infirmary visits

Recommended Tier: These workspaces are offered as standard extensions to the core package. Most medium-to-large institutions benefit from activating 3-5 recommended workspaces based on their operational model.

The **Parent Portal** is the most commonly adopted recommended workspace — it transforms parent engagement by providing real-time access to attendance, grades, fee status, and school announcements.

Transport and **Library** integrate directly with the Payments workspace, enabling automatic fee deductions for bus passes and late book returns.

Conditional Workspaces for Specialized Institutions

CONDITIONAL WORKSPACES

Hostel

Room allocation, mess management, attendance, and disciplinary records for boarding schools.

Activates when the institution operates residential facilities.

Alumni

Network directory, donation campaigns, reunion events, and mentorship programs.

Activates for institutions with established graduate networks.

Placement

Company tie-ups, drive scheduling, student eligibility tracking, and offer management.

Activates for higher secondary and collegiate institutions.

Research & Development

Project tracking, grant management, publications, and collaboration tools.

Activates for research-focused institutions and universities.

Platform Workspaces Enable System Governance

PLATFORM WORKSPACES

Team Owner

Super Admin workspace for tenant provisioning, global billing, security policies, and platform configuration.

User Management

Role creation, permission matrices, bulk user imports, and comprehensive audit logging.

Course Coordinator

Cross-institution curriculum alignment and board standard management for CBSE, ICSE, and IGCSE.

YANTRA

Central AI orchestration layer connecting all workspaces to intelligence services and predictive analytics.

Governance Layer: Platform workspaces are invisible to end users but essential to system integrity. They form the control plane that keeps the entire VID ecosystem secure, consistent, and scalable.

The **Team Owner** workspace is the only interface with access to tenant creation and deletion. It manages subscription tiers, payment methods, and global security policies.

User Management provides a single pane for role engineering. Administrators can create custom roles, define permission matrices, and export audit trails for compliance reviews.

Course Coordinator ensures that curriculum changes propagate correctly across institutions following the same board standard without manual duplication.

YANTRA AI Engine Powers Intelligent School Operations

YANTRA AI ENGINE

Chatbot

Natural language queries for students, parents, and staff. Instant answers from institutional data — timetables, fees, results, and policies.

Voice Agent

Hands-free interaction for teachers and administrators. Execute commands, retrieve data, and manage tasks via voice.

Face Recognition

Automated, fraud-proof attendance marking with real-time sync. Eliminates proxy attendance and manual roll calls.

Cross-Workspace AI

AI correlates data across Academic, Examination, and Student workspaces to predict at-risk students and optimize operations.

YANTRA in Action: The AI engine doesn't just automate — it predicts and recommends. By analyzing patterns across workspaces, YANTRA identifies students showing early warning signs of academic decline.

For example, when a student's attendance drops (Academic workspace), assignment submission rate falls (Student workspace), and exam scores trend downward (Examination workspace), YANTRA flags the student as "at-risk" and recommends intervention to the Principal Dashboard.

The **Face Recognition Attendance** system processes biometric data locally for privacy, syncing only anonymized attendance records to the cloud. It integrates directly with the Academic and HRMS workspaces.

Five Architectural Pillars Define the VID Platform

KEY FEATURES

Multi-Tenant

One platform, unlimited institutions, complete data isolation between tenants.

Modular Design

Enable only needed workspaces — no bloat, no unused features cluttering the interface.

Scalable

Cloud-native, containerized services with auto-scaling for growing user bases.

Secure Isolation

Row-Level Security, encrypted connections, and comprehensive audit trails.

Plug & Play

New modules integrate without disrupting existing operations or requiring downtime.

Architecture Summary: VID's five pillars are not marketing labels — they are engineering constraints. Every design decision must satisfy all five criteria before it enters the codebase.

Multi-tenancy is enforced at the database layer through PostgreSQL RLS, not just application logic. This means even a direct database connection cannot breach tenant boundaries.

Modularity extends to the frontend: each workspace ships its own UI components, reducing bundle size and enabling independent deployment.

Scalability is achieved through horizontal pod autoscaling in Kubernetes. During admission season, the Admission workspace scales out independently while other workspaces remain unaffected.

Cloud-Native Infrastructure Built for Scale

INFRASTRUCTURE DESIGN

Backend

API-based microservices architecture built with containerized services. Each workspace runs as an independent service with its own deployment lifecycle. Docker and Kubernetes orchestration enable horizontal scaling, rolling updates, and zero-downtime deployments.

Frontend

Workspace-specific UI components with responsive design. Progressive Web App ready for mobile and tablet access without app store dependencies. Component isolation ensures a bug in one workspace UI cannot crash the entire application.

Database

PostgreSQL with Row-Level Security policies enforcing tenant isolation at the database layer. Automated backups, point-in-time recovery, and read replicas for performance.
RLS means even direct SQL access respects tenant boundaries — impossible to bypass.

Auth & DevOps

JWT-based authentication with refresh tokens, MFA support, and SSO integration readiness. CI/CD pipelines with automated testing and blue-green deployments. Infrastructure-as-Code ensures reproducible environments across dev, staging, and production.

VID Delivers Measurable Value Across Five Dimensions

KEY BENEFITS

Scalability

Add institutions, users, and workspaces without performance degradation or infrastructure overhaul.

Security

Tenant-isolated data, encrypted transit and rest, role-based access, and complete audit logging.

Cost Efficiency

Shared infrastructure reduces per-institution IT costs by 60%+ compared to standalone ERP deployments.

Maintainability

Modular architecture means updates to one workspace never affect others — isolated failure domains.

Future Ready

AI-native design, API-first architecture, mobile-ready, and marketplace integration ready.

Value Proposition: VID transforms education IT from a cost center into a strategic enabler. The multi-tenant model means infrastructure costs are shared across all institutions, while each school pays only for the workspaces they use.

Scalability ensures that a school group can grow from 2 to 200 campuses without replacing their ERP. **Security** guarantees that a data breach in one tenant cannot compromise another.

Cost efficiency comes from eliminating redundant infrastructure, reducing manual processes, and consolidating vendor relationships. **Maintainability** means your IT team spends less time on emergency patches and more time on innovation.

Most importantly, **future readiness** means the platform you deploy today will support the AI, IoT, and mobile technologies of tomorrow without a rip-and-replace migration.

Roadmap Positions VID as the Education Operating System of Tomorrow

FUTURE ENHANCEMENTS

01

AI Dashboards

Predictive analytics for enrollment forecasting, performance trends, and resource planning. Machine learning models trained on institutional data to recommend optimal class sizes, faculty allocation, and budget distribution.

02

IoT Attendance

Biometric and RFID integration for seamless attendance across campuses. Gate-based check-in, classroom beacons, and wearable devices for automatic presence detection without manual intervention.

03

Mobile App

Native iOS and Android applications for students, parents, and staff with offline support. Push notifications for announcements, fee reminders, and emergency alerts. Full workspace functionality in pocket-sized form.

04

Marketplace Integrations

Connect with payment gateways, LMS providers, video conferencing tools, and government education portals. Open API ecosystem enabling third-party developers to build and sell VID-compatible workspaces.

VID

The Education Operating System

Intelligent, scalable, and modular ERP platform designed for the future of education

From a single classroom to a global network of institutions, VID grows with you. Multi-tenant architecture, modular workspaces, and AI-powered intelligence — everything schools need to thrive in the digital age.

NEXT STEPS

Step 1

Schedule a personalized demo tailored to your institution's board and size.

Step 2

Pilot VID with a single campus or department to validate fit and ROI.

Step 3

Scale across your institution network with full implementation support.

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Tagline: Empowering Education Through Technology